

MANAN KAPOOR

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EDUCATION

Thapar Institute of Engineering & Technology
Bachelor of Engineering in Computer Engineering
CGPA: 8.07 / 10.00

Punjab, India
2024 – 2028

EXPERIENCE

Research Intern2025 – Present

Thapar Institute of Engineering & Technology, Patiala, India
Python — NLP — Dataset Engineering — LLM Fine-Tuning (QLoRA) — Teacher-Model Supervision — Controlled Text Generation — BLEU/BERTScore Evaluation — Gemini API — Unsloth — HuggingFace Transformers

- Designed and published **PLC-Rewrite** — the first multi-task Punjabi instruction dataset for controlled text rewriting — aggregating **16.43M tokens** from **600+** curated web sources; engineered a **Gemini 2.0 Flash** teacher-model supervision pipeline generating **91,000+ instruction-conditioned samples** across **6 rewriting tasks**; applied task-specific structural validation eliminating **~37%** corrupt samples and achieving **100% instruction-compliance** across 3,000 evaluated samples with BERTScore ≥ 0.929 on every task.
- Fine-tuned **LLaMA 3.1 8B** on PLC-Rewrite using **QLoRA** (4-bit NF4, $r=64$, $\alpha=128$, **167M trainable parameters**); reduced training loss **0.55** $\rightarrow$ **0.34** over 3 epochs — fine-tuned model produces consistent **Gurmukhi script output** across all 6 tasks while the base model fails entirely (defaulting to Devanagari or character-level noise), establishing PLC-Rewrite as an effective supervised training resource for low-resource Punjabi NLP.

PROJECTS

KV-Paged Inference System for Transformer Modelsgithub.com/manankapoor23/KV-Paged-Attention

Python — PyTorch — Transformers — ML Systems — OS Concepts

- Architected a paged Key-Value (KV) cache system for transformer inference, decoupling logical token positions from physical memory to achieve **40%+ memory efficiency gains** in long-context decoding.
- Engineered allocator-backed KV pages with page tables, reference counting, prefix reuse, and copy-on-write semantics to optimize memory sharing across concurrent inference requests, reducing redundant allocations by **~35%**.
- Validated paged attention via numerical equivalence testing against naive attention, profiling memory growth and compute behavior during autoregressive generation with **<0.1% precision delta**.

Natural Language to SQL Query Systemgithub.com/manankapoor23/NL-SQL

Python — LangChain — LLMs — SQL — Slack API — Prompt Engineering

- Built an **LLM-powered NL $\rightarrow$ SQL pipeline** that converts natural language questions into executable SQL queries, enabling intuitive querying of relational databases directly from Slack.
- Developed a **Slack-integrated data assistant** using LangChain that processes slash-command queries, generates schema-aware SQL, executes it against a live database, and returns structured results in-channel.

TECHNICAL SKILLS

Languages	Python, C++, JavaScript, R
ML & AI	Machine Learning, Deep Learning, Transformers, LLMs, Generative AI, RAG, Prompt Engineering, Fine-Tuning (QLoRA), Instruction Tuning
Libraries	PyTorch, HuggingFace Transformers, Unsloth, LangChain, Pandas, NumPy, Scikit-learn, Matplotlib, OpenAI SDK, NLTK
Data Engineering	Dataset Curation, JSONL Pipelines, Deduplication, Schema Validation, Feature Engineering, Cross-Validation, Hyperparameter Tuning
Databases	SQL, PostgreSQL, Vector DBs (FAISS, Chroma)
Tools	Git/GitHub, Docker, VS Code, Jupyter, RStudio
CS Fundamentals	DSA, OOP, Operating Systems, DBMS, Computer Networks, Probability & Statistics